

# Speaker Transcript

## Antenna Theory and Design - Chapter 1

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Week 1

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## Slide 1: Title Slide

Welcome to Week 1. In this lecture, we begin with the big picture of antenna engineering. The goal is not only to define an antenna, but to understand why antennas are central to wireless communication, radar, sensing, and array signal processing. Chapter 1 of Stutzman and Thiele gives us the conceptual foundation: antennas as interfaces between guided waves and free-space waves, the physical mechanism of radiation, the most important performance parameters, and the four major antenna families.

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## Slide 2: Learning objectives

By the end of the lecture, students should be able to explain what an antenna does, why physical size must be compared with wavelength, and how the major parameters describe antenna behavior. The numerical example will connect the qualitative definitions to engineering quantities such as wavelength, input impedance, return loss, VSWR, gain, and bandwidth.

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## Slide 3: Chapter 1 roadmap

The roadmap follows the structure of Chapter 1. We first build the historical and physical motivation. Then we define antennas as transducers between guided and radiated waves. After that, we discuss the physics of radiation and the practical parameters used to evaluate antennas. Finally, we classify antennas and close with a numerical example.

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## Slide 4: Historical motivation: from Maxwell to modern wireless

Antenna engineering is rooted in the history of electromagnetics. Maxwell provided the theoretical prediction of electromagnetic waves. Hertz experimentally verified electromagnetic wave propagation and built early antennas. Marconi transformed wireless telegraphy into practical long-distance communication. Later, radar and microwave technology during World War II made horn, reflector, and array antennas central to engineering. In the modern era, cellular systems, satellites, GPS, WLAN, and radar all rely on antenna design.

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## Slide 5: What is an antenna?

The most important definition is that an antenna is a transducer between a guided wave and a free-space wave. On transmit, the energy begins as a signal on a transmission line. The antenna

allows that energy to detach from the structure and propagate through space. On receive, the process reverses. An incoming electromagnetic wave induces currents and voltages at the antenna terminals.

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## **Slide 6: Transmission line versus antenna**

A transmission line and an antenna have opposite design goals. A transmission line tries to keep the fields bound to the structure. An antenna tries to launch the fields into space. This contrast is one of the most useful ways to understand the antenna concept. In practice, an antenna can be viewed as a controlled transition from a guided-wave structure to a radiating structure.

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## **Slide 7: Wavelength is the design scale**

The wavelength is the natural ruler of antenna engineering. A small object in ordinary dimensions may be electrically large if the wavelength is small. Conversely, a physically large object may be electrically small at very low frequencies. This is why antenna design changes so strongly with frequency. At 2.4 GHz, the wavelength is only 12.5 centimeters, so practical antennas can be quite compact.

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## **Slide 8: When do we use antennas?**

Antennas are essential whenever the system cannot be physically connected by a cable or when broadcasting is required. They are also central in radar, where the antenna determines where energy is transmitted and from what directions signals are received. The cable-versus-radio decision is not only about loss. It also involves cost, mobility, reliability, and system architecture.

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## **Slide 9: Physical mechanism of radiation**

Radiation begins with acceleration of charge. In an antenna, the RF generator causes current and charge distributions to oscillate. These time-varying currents and charges produce time-varying electric and magnetic fields. The field disturbance propagates away from the antenna. The radiated fields decrease approximately as one over distance, while power density decreases as one over distance squared because the power spreads over a larger spherical surface.

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## Slide 10: From open transmission line to dipole

A very useful physical model is to start from an open transmission line and bend the two conductors outward. In the original line, the opposite currents and confined field structure keep the energy guided. When the conductors are opened outward, the current distribution can radiate into free space. This is one of the simplest ways to understand how a half-wave dipole evolves from a transmission-line viewpoint.

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## Slide 11: Antenna performance parameters

Chapter 1 introduces the main language of antenna engineering. Radiation pattern tells us the directional behavior. Directivity tells us how concentrated the radiation is. Gain includes efficiency. Polarization describes the electric-field orientation. Input impedance determines how effectively the transmitter or receiver connects to the antenna. Bandwidth defines the frequency range of acceptable performance. Scanning is important for radar and phased arrays.

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## Slide 12: Radiation pattern and directivity

The radiation pattern is often the most important antenna characteristic. It tells us where the antenna sends or receives energy. Directivity is computed from the pattern and quantifies how much radiation is concentrated in the preferred direction. Directivity does not include losses. A lossless antenna and a lossy antenna can have the same directivity if their pattern shapes are the same.

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## Slide 13: Gain, efficiency, and power interpretation

Gain combines directivity with radiation efficiency. Efficiency accounts for power lost in conductors, dielectrics, surface currents, or nearby structures. If an antenna accepts one watt and has ninety percent radiation efficiency, then only zero point nine watts is radiated. This is why gain is usually more meaningful than directivity in system link budgets.

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## Slide 14: Polarization, impedance, bandwidth, and scanning

Polarization matters because a mismatch between transmitting and receiving polarization can reduce received power. Input impedance matters because the antenna must accept power from a practical RF system, often fifty ohms. Bandwidth should always be defined using a criterion, such

as return loss less than minus ten decibels, gain variation below a specified tolerance, or stable pattern shape. Scanning becomes central when we move to phased arrays and radar systems.

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## **Slide 15: The four basic antenna types**

Chapter 1 organizes antennas into four major families. Electrically small antennas are much smaller than the wavelength and are often difficult to match efficiently. Resonant antennas operate well over narrow bands and include dipoles, patches, and Yagi antennas. Broadband antennas maintain acceptable behavior over wide frequency ranges. Aperture antennas use a physical opening or reflecting structure and are important in microwave, radar, and satellite applications.

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## **Slide 16: Electrically small and resonant antennas**

Electrically small antennas are attractive when the available physical space is limited, but they create matching and efficiency challenges. Resonant antennas are more common in introductory design because their dimensions are tied to simple fractions of wavelength. The half-wave dipole is the classical example, and the rectangular microstrip patch is another important resonant antenna in modern RF systems.

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## **Slide 17: Broadband and aperture antennas**

Broadband antennas are designed so that key parameters remain acceptable over a large frequency interval. An important concept is the active region, where the portion of the antenna responsible for radiation changes with frequency. Aperture antennas are typically used at UHF, microwave, and millimeter-wave frequencies. Horns and reflectors can provide high gain because they control the spatial distribution of the aperture field.

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## **Slide 18: Choosing antenna type by gain target**

Chapter 1 provides a useful engineering rule of thumb. Low gain can be achieved with simple wire antennas. Moderate gain may come from patches or broadband planar antennas. Higher gain requires antennas such as Yagi-Uda, helices, or log-periodic arrays. Very high gain usually requires an aperture antenna or an array. This is a practical way to begin antenna selection before detailed design.

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## Slide 19: Numerical example: half-wave dipole at 2.4 GHz

Now we turn the definitions into numbers. A half-wave dipole at 2.4 GHz is a convenient example because it is simple, physically small enough to visualize, and representative of resonant antennas. We assume it is vertical, center-fed, resonant, and connected to a 50-ohm system.

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## Slide 20: Example: wavelength and physical length

At 2.4 GHz, the free-space wavelength is 12.5 centimeters. A half-wave dipole would ideally be about 6.25 centimeters long, but practical wire dipoles are often slightly shorter because of end effects and conductor diameter. A reasonable practical value is about 0.48 wavelength, or about 6 centimeters.

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## Slide 21: Example: pattern, directivity, and gain

For a vertical half-wave dipole, the maximum radiation is broadside to the antenna, in the horizontal plane. There are nulls along the antenna axis. The pattern resembles a doughnut. Its half-power beamwidth is roughly 78 degrees. The directivity is approximately 1.64, or 2.15 dBi. With 90 percent efficiency, the gain becomes 1.476 as a power ratio, or about 1.69 dBi.

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## Slide 22: Example: polarization and input impedance

A vertical dipole is linearly vertically polarized. If the receive antenna has the same polarization, the polarization match is good. If it is tilted, the received power is reduced by the polarization loss factor  $\cos^2 \psi$ . At resonance, the input impedance of a thin half-wave dipole is approximately 73 ohms with zero reactance. This is close to, but not exactly equal to, a 50-ohm RF system.

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## Slide 23: Example: mismatch calculations

The mismatch between 73 ohms and 50 ohms gives a reflection coefficient magnitude of about 0.187. The return loss is therefore about 14.6 dB, which is a reasonably good match for many RF systems. The VSWR is about 1.46. If one watt is incident on the antenna, about 96.5 percent is accepted and about 3.5 percent is reflected. The mismatch loss is small, about 0.15 dB.

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## Slide 24: Example: bandwidth and summary

Bandwidth must be defined with a specific criterion. Here, we use the common impedance criterion  $S_{11}$  less than minus 10 dB. If the antenna satisfies this from 2.34 to 2.46 GHz, the absolute bandwidth is 120 MHz and the fractional bandwidth is 5 percent. The summary table collects the main numbers for the 2.4 GHz half-wave dipole.

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## Slide 25: Connection to radar, arrays, and DOA

This is the direct connection to radar and direction-of-arrival estimation. In an ideal signal-processing simulation, we often use point sensors with identical isotropic responses. In real hardware, every antenna element has a pattern, gain, polarization, impedance, and coupling behavior. The measured array response is the product of element behavior and the array factor. This affects SNR, angular bias, array calibration, and the accuracy of algorithms such as MUSIC and beamforming.

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## Slide 26: Final takeaways

The main message of Chapter 1 is that an antenna is a physical-circuit-spatial system. It cannot be judged by a single number. We must consider radiation pattern, directivity, gain, polarization, impedance, bandwidth, scanning, and system constraints together. For students moving toward radar, array processing, or RF hardware, this chapter is the foundation for connecting electromagnetic physics to signal processing and system design.

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## Slide 27: References

This presentation is based primarily on Chapter 1 of Stutzman and Thiele's Antenna Theory and Design. The numerical example uses standard half-wave dipole approximations that are commonly introduced in antenna fundamentals.

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